Foundation Models for Metabolic Engineering

Genesis Mission Phase I proposal, redacted transcription

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A transcription of the Phase I proposal submitted to the Genesis Mission request for applications on 2026-04-23, under focus area 2-B, scaling the biotechnology revolution, genotype to phenotype. Person names, telephone numbers, email and street addresses, the RFA number, the named industry partners and every budget figure are replaced by a gray [redacted: what was removed] marker. Institutions, facilities, the scientific content, the seven go/no-go metrics of Table 3 and the bibliography are kept as submitted, so this copy can be read for what the award commits the project to, without carrying the administrative data of the submission.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 Cover block ✗

Project title Foundation Models for Metabolic Engineering

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RFA number [redacted: RFA number]

Primary challenge and focus area (RFA Section III) 2-B Scaling the Biotechnology Revolution, Genotype to Phenotype

Table 1: Senior/key personnel for the lead institution and all partner institutions. Names are redacted; the column records one named person per institution, as submitted.

Institution name	Senior/key personnel including the Director
University of Illinois Urbana-Champaign (lead)	[redacted: lead PI]
Lawrence Berkeley National Lab	[redacted: partner PI]
Argonne National Lab	[redacted: partner PI]

Table 2: Summary budget information for the lead institution and all partner institutions, Year 1. Every figure is redacted; the submitted table carried one amount per row and a total.

Institution	Year 1 budget
University of Illinois Urbana-Champaign (lead)	[redacted: amount]
Lawrence Berkeley National Lab	[redacted: amount]
Argonne National Lab	[redacted: amount]
Total	[redacted: amount]

2 Background and introduction ✗

Metabolic engineering has been increasingly used to produce chemicals, fuels, and materials[1]. Developing AI foundation models for metabolic engineering is critical for addressing the complexity and scale of biological design problems. Engineering microbial metabolism requires navigating an enormous combinatorial space of genes, enzymes, regulatory interactions, and environmental conditions. Traditional modeling and machine-learning approaches are typically trained for narrow tasks and rely on limited labeled datasets, making them difficult to generalize across organisms and pathways. In contrast, foundation models – large AI models pretrained on diverse biological datasets such as genomic sequences, metabolic networks, protein structures, and multi-omics measurements – can learn transferable representations of biological systems and be adapted to many downstream tasks with relatively little additional data[2]. In essence, by integrating heterogeneous biological data and capturing complex dependencies within cellular systems, AI foundation models link genotype and phenotype (Focus area 2B) and can guide pathway discovery, retrosynthetic planning, and strain optimization while reducing costly experimental trial-and-error[3]. Ultimately, they could transform metabolic engineering into a more predictive and scalable discipline, enabling rapid development of sustainable biomanufacturing platforms for renewable chemicals, fuels, and materials (AI advantages).

The three partner groups (the lead university group, the LBNL group, and the ANL group) are pioneers in developing AI models for metabolic engineering. For example, the lead group developed an AI-driven automated workflow for pathway engineering in 2019[4] while the LBNL group developed the Automated Recommendation Tool (ART), a tool that leverages machine learning and probabilistic modeling techniques to guide metabolic engineering in a systematic fashion in 2020[5]. Moreover, the ANL group developed GenSLM, a DNA large language model (LLM), which was able to predict novel variants in the newly emerging coronavirus genomes (SARS-CoV-2) in 2023. Recently, in an unpublished study, the lead group developed TorchCell, a framework to facilitate building metabolic engineering foundation model (MEFM) with specific applications in *Saccharomyces cerevisiae*, a preeminent industrial platform organism (Fig. 1). Yeast genes are embedded using various pretrained DNA and protein embedding models, and protein-protein interactions, regulatory networks, and metabolism are then used to build adjacency matrices among genes, reactions, and metabolites

demonstrating multimodal input representation of metabolic engineering experiments. We select gene-gene epistasis, commonly referred to as gene-gene interactions (GGI), as our supervised prediction label since it is a well-defined representation of the nonlinear complexity of biological systems that makes it difficult to reliably engineer commercially viable strains. The TorchCell Cell Graph Transformer (CGT) model is trained on a dataset queried from the TorchCell Database, which aggregates multiple original datasets (Fig. 1a–c). It achieves a Pearson correlation of 0.454 ± 0.006 and a Spearman correlation of 0.421 ± 0.004, with increased noise at the extremes of the GGI range, as expected (Fig. 1d). We demonstrate state-of-the-art performance from benchmarking against flux balance analysis (FBA), the common mechanistic, constraint-based modeling method, and two recent deep learning frameworks DCell and DANGO[9–11] (Fig. 1e). Notably, many foundation models have been developed for protein engineering; however, no foundation model has been developed for metabolic engineering. This is largely due to the fragmentation of phenotypic data across diverse experimental datasets, where individual studies are typically too small to support deep learning approaches, preventing the field from understanding how models scale across metabolic engineering data types. TorchCell fills this gap by aggregating genotype, phenotype, and environmental data relevant to metabolic engineering.

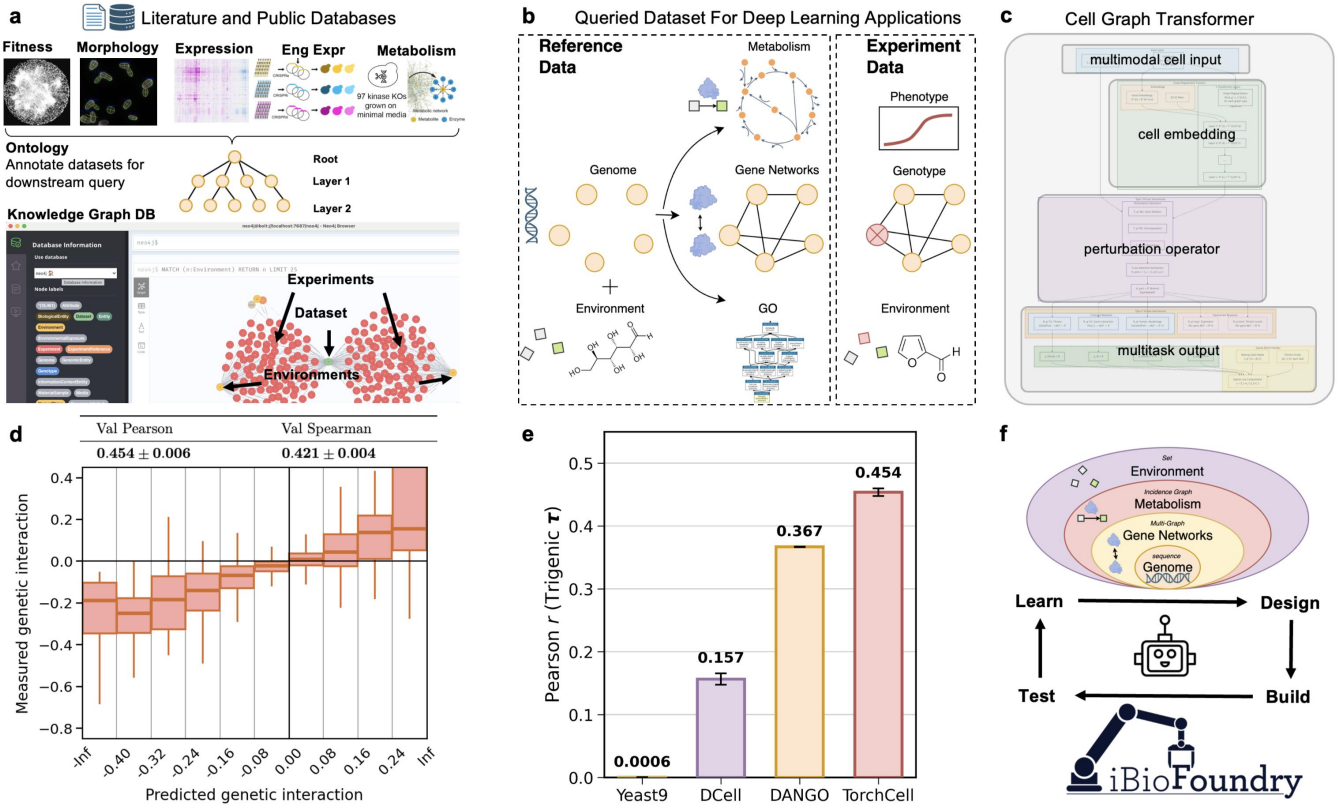


Figure 1: TorchCell database construction and *S. cerevisiae* application predicting cellular epistasis. **a**, Literature and public databases are annotated with TorchCell's experiment ontology, enabling coherent querying across all data in the knowledge graph database. Experiments are added to the TorchCell Graph database hosted by National Center for Supercomputing Applications (NCSA), which can be queried to construct datasets large enough to enable DL applications. **b**, TorchCell's minimal requirements for genomic representation are a full genome sequence and a genome annotation file, but layers of additional graphical knowledge like regulatory networks and metabolism can be added if annotations exist. Full representations of experiments are often highly repetitive, so TorchCell compresses them into perturbation operators over a reference state, facilitating training on millions of strains. **c**, Cell Graph Transformer (CGT) takes multimodal cell inputs, embeds them into cell latent space via a perturbation operator as described in recent virtual cell literature, and the perturbed cell embeddings are decoded into different supervised labels, in this case, GGI[7–8]. **d**, TorchCell CGT validation on GGI. **e**, TorchCell CGT achieves state-of-the-art performance in predicting trigenic GGI in yeast compared to Yeast9 FBA, DCell, and Dango. **f**, TorchCell can be used to learn from previous strain experiments and help design the next batch of strains for experimentation in autonomous labs like the NSF iBioFoundry at UIUC.

3 Project objectives ✗

Building upon our previous accomplishments, we propose to develop a metabolic engineering foundation models (MEFMs) through the following specific aims:

1. Extend TorchCell to facilitate the development of MEFMs by expanding the TorchCell Database and improving metabolic engineering model architecture. We will extend the current TorchCell CGT to other organisms by

finetuning and retraining on perturb-seq data[12] collected from industrial platform organisms.

2. Integrate an improved genome large language model (LLM) based on GenSLM for metabolic engineering (HiGenLM) finetuned on data from TorchCell Database.

3. Use our biofoundries, including the iBioFAB, NSF iBioFoundry, SynBioAgent agentic AI system at the UIUC, the OPAL autonomous laboratory platform, and the IDeA co-scientist infrastructure at the ANL, to validate AI model predictions and generate large amounts of data in a high-throughput manner for finetuning and training these models.

In Phase I (first 9 months), our main goal is to demonstrate proof of concept by unifying the aforementioned AI models into an MEFM and establishing computational and wet-lab experimental pipelines for scale-up studies in Phase II. In Phase II (next three years), we aim to include all 100–250 industrially relevant and emerging organisms used for metabolic engineering in the TorchCell Database and host metabolic engineering benchmark datasets for the metabolic engineering community. We will also collect data on at least twenty organisms widely used in the field of metabolic engineering using our agentically driven biofoundries, and we will use TorchCell to train metabolic engineering foundation models on at least 1000 GPUs in the soon-to-be-established Equinox and Solstice systems at ANL. In addition, we will collaborate with [redacted: two industry partners] to validate our MEFMs using their platform organisms, and with [redacted: an instrument vendor] to enhance the technical capabilities of our biofoundries for experimental validation and data generation (see letters of support). Note that this project will not involve the construction, alteration, maintenance and/or repair of public infrastructure in the United States.

4 Proposed research and methods ✗

4.1 Aim 1. Extend TorchCell for generalizable foundational models in metabolic engineering ✗

To expand TorchCell to other important industrial organisms, we will start with the well-studied bacterial strain *Escherichia coli* and expand to 10 additional gram-negative bacteria and 10 additional yeasts. We will then demonstrate applications on *Pseudomonas putida*, which has recently been optimized for isoprenol production[11], an important aviation fuel precursor. In addition, this case study will help evaluate recommendations provided by TorchCell in DBTL loops (Fig. 2a)[11].

Although large foundation models have shown some promising results in mammalian cells, systematic benchmarking indicates that transformer-based perturbation models do not consistently outperform simple linear baselines[13]. This gap can be filled by better mechanistic representations grounded in physical constraints, as demonstrated by Merzbacher et al. in their application of deep learning to flux cone geometry to predict metabolite production[14]. Since TorchCell already facilitates the inclusion of metabolism, we intend to further elaborate its description to ensure deep learning cell representations are well grounded in physical constraints (Fig. 2c). *Successful applications in organisms with comparatively less data would demonstrate the successful implementation of MEFMs.* (Underlined in the submission; italic here because an underline cannot break across lines.)

4.2 Aim 2. Integrate an improved genome LLM based on GenSLM for metabolic engineering (HiGenLM) with TorchCell ✗

GenSLM was one of the first codon-based LLM models to enable genome-scale predictions and to predict novel variants in the newly emerging coronavirus genomes (SARS-CoV-2)[6]. We are now developing a phenotypically informed DNA LLM that leverages GenSLM and performs post-training with phenotypic data to make effective phenotype predictions. Our new version of GenSLM, HiGenLM, focuses on automatically learning the context of nucleotides, genes, contigs, and operons in the genome, using a nested, multi-resolution transformer architecture pre-trained on genomic sequences from thousands of microbial and eukaryotic genomes. We hypothesize that this hierarchical context learning is critical to add phenotypic awareness. At its core, HiGenLM decomposes the genome into a hierarchy of overlapping biological units, codons, genes, operons, regulatory modules, and chromosome segments, and learns representations at each scale through a combination of masked language modeling and cross-scale contrastive objectives. The result is a set of scale-aware genomic embeddings that capture both the fine-grained biochemical specificity of individual protein-coding sequences and the broader genomic context in which they operate. Critically, because HiGenLM is pre-trained across phylogenetically diverse organisms, its representations encode evolutionary conservation signals that genome-scale models entirely lack: residues under purifying selection, syntenic gene neighborhoods, and cross-species regulatory grammar are all implicitly encoded in the embedding geometry. Currently, we are training HiGenLM with phenotypic data by leveraging Directed Preference Optimization (DPO)[15], which will help us tackle the disparity of phenotypic

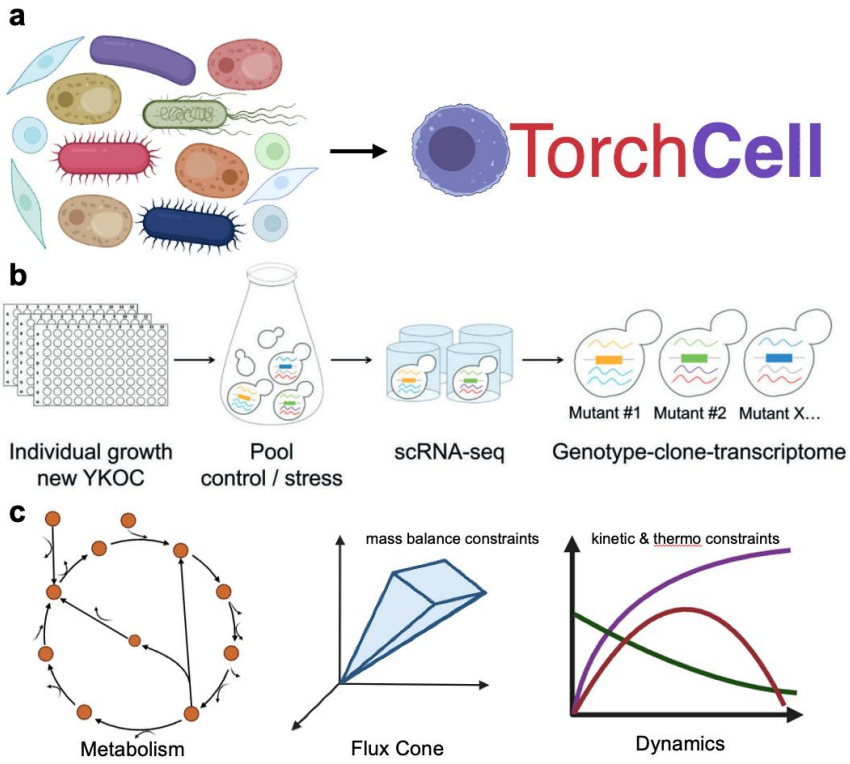


Figure 2: The figure as embedded in the submission, which carries no caption of its own. **a**, organisms entering TorchCell. **b**, the perturb-seq style atlas workflow: individual growth of a new yeast knockout collection, pooling under control and stress conditions, scRNA-seq, and the genotype-clone-transcriptome readout. **c**, the metabolic grounding layers: the metabolic network, the flux cone under mass balance constraints, and dynamics under kinetic and thermodynamic constraints.

and genotypic data availability. DPO works by directly updating the LLM using a loss function that favors preferred datasets (e.g., higher production phenotypes). An important part of this process will be augmenting phenotypic data to meet post-training needs, which will be done by extrapolating from current data and by supplementing phenotypic data from the TorchCell database.

We will attempt to integrate HiGenLM with TorchCell. HiGenLM gene-level embeddings can replace or augment the sequence features currently used in TorchCell's supervised models for fitness, gene interaction, morphology, and expression prediction. By providing each gene node in TorchCell's graph database with a HiGenLM-derived embedding giving TorchCell CGT models access to evolutionary and functional priors that sharply reduce the effective sample complexity for multi-phenotype prediction. Preliminary evidence from analogous integrations using GenSLM applied to *E. coli* pathway engineering suggests that this embedding transfer can improve phenotype prediction accuracy by 15–30% in low-data regimes, precisely the regime most relevant to metabolic engineering strain development. Notably, HiGenLM's cross-species pre-training provides a shared embedding space in which gene function and regulatory logic are transferable across phylogenetic distance, enabling TorchCell-style phenotype models to be rapidly adapted to new hosts by grounding their gene representations in HiGenLM's universal genomic language.

4.3 Aim 3. Use biofoundries to validate the predictions made by the proposed AI models and generate large amounts of data in a high-throughput manner for finetuning of these models ✗

Metabolic engineering relies on genetic manipulation to redirect flux within organisms, necessitating data collection across perturbations of the genome and environment to construct effective metabolic engineering foundation models. We propose to collect perturb-seq style single-cell atlas[12] for non-model organisms relevant to industrial biotechnology. Data will be used to evaluate foundation models built on model organisms (e.g., *S. cerevisiae* and *E. coli*) and applied to non-model organisms, and to assess model improvements using high-throughput perturbation data. Selected strains will also be screened with metabolically relevant metabolites using EchoMS to link high-throughput sequencing data to more expensive metabolite quantification using previously mentioned facilities.

To validate the gene target predictions made by the MEFMs, we will focus on model organisms *S. cerevisiae* and *E. coli* for which we have established automated workflows for gene knockouts and overexpression on our iBioFAB and NSF iBioFoundry (unpublished data). Building upon our prior studies[4, 16–17], we will select β-carotene as a target compound for *S. cerevisiae*, and lycopene as a target compound for *E. coli*. These organisms and relevant assays have

been well established in the lead group and will help evaluate MEFM predictions for the transfer of pathways to new hosts.

5 Milestones in the nine months ✗

This proposal is structured to establish milestones towards building MEFMs with a 6-month decision point aligned with the go/no-go metrics (Table 3). These goals collectively organize the research program into stages that help address gaps in the development of MEFMs. These include scaling data across organisms and genotypes, designing a metabolic representation module, and demonstrating the real-world use of these models by designing strains. Following the development of the first MEFMs, these models will be used in metabolic engineering applications to generate data from recommended strains, which will be fed back into the models to complete the first continuous improvement loop for MEFMs.

In the first 6 months, there will be a computational track (i–iii, v) adding new data to the current TorchCell database, developing TorchCell CGT models for *E. coli*, and grounding virtual cell models to respect mass balance and thermodynamics; and the experimental track (iv) where a protocol will be developed for an RNA-barcoded YKOC experiment in *I. orientalis* where the subsequent atlas will be published. Improved TorchCell CGT models for *S. cerevisiae* and *E. coli* will be finetuned (v) using published data for *P. putida* and preliminary in-house data for *I. orientalis*. These models will be used for strain design recommendations for 3-HP production in *I. orientalis* and isoprenol production in *P. putida*. At the 6-month decision point, success will be evaluated using quantitative thresholds aligned with the decision matrix (Table 3). Achieving these criteria (i–v) will demonstrate the scaling of TorchCell's database and experimental ontology across diverse organisms and phenotypes, and that physically grounded MEFMs can be used for strain design across eukaryotes and prokaryotes.

Table 3: Six-month go/no-go decision metrics and AI advantage criteria, as submitted (the proposal's Table 1).

Metric	Go	AI advantage	Months
i) TorchCell Database Expansion	10 bacteria and 10 yeasts plus key organisms *E. coli*, *P. putida*, and *I. orientalis* added to the TorchCell database with $\geq$ 1,000 experiment instances and $\geq$ 3 phenotypes each.	Shared experimental ontology with TorchCell dataloaders converts heterogeneous data $\geq$ 20 organisms into benchmark-ready DL datasets for building MEFMs.	1–6
ii) TorchCell CGT *E. coli*	TorchCell CGT benchmarked on *E. coli* demonstrating phenotype prediction exceeding FBA and linear baselines $\geq$ 20%.	$\geq$ 20% improvement over FBA and linear baselines demonstrates multimodal representations capture biological nonlinearities beyond shallower methods.	2–4
iii) Metabolic Grounding	Addition of metabolism module to TorchCell CGT so that $\geq$ 95% of top 100 strains don't violate mass balance or ΔG feasibility.	Physical constraints filter implausible strain designs and are independent of training-data scale, critical for non-model organisms.	1–6
iv) *I. orientalis* Atlas	Developed protocol for *I. orientalis* RNA-barcoded YKOC across $\geq$ 3 environmental conditions yielding $\geq$ 5,000 perturbation-condition pairs.	First genome-×-environment perturbation atlas in *I. orientalis* establishing a template for scaling across organisms and providing highly dense training signal for this organism of interest.	1–9
v) Finetune TorchCell CGT	TorchCell CGT finetuned from *S. cerevisiae* to *I. orientalis*, and from *E. coli* to *P. putida* matches from scratch baseline Pearson with $\geq$ 5× fewer target labels.	5× sample efficiency on cross-organism transfer quantifies the returns of adding new host organism data and provides scaling evidence for expanding MEFMs.	4–6
vi) TorchCell CGT Foundation Model Training Expansion	TorchCell CGT models trained across $\geq$ 10 yeasts and 10 $\geq$ bacteria with leave-one-organism-out Pearson $\geq$ 0.3 and reported transfer exponents with 95% CIs.	Scaling exponents with uncertainty quantify the return per added organism and phenotype, providing measured scaling behavior for MEFM training.	6–9
vii) TorchCell CGT Metabolic Engineering Applications	$\geq$ 1 strain showing ($p < 0.05$, $n \geq 3$) titer improvement over base strain for 3-HP production in *I. orientalis* and isoprenol production in *P. putida*, verified by high-throughput mass spec.	Automated biofoundries scale experimental DBTL throughput while feeding results back to MEFM, demonstrating computational and experimental scaling for future strain engineering campaigns.	6–9

During months 6–9 TorchCell models will be retrained (vi) on data within the same domain of life to lay the blueprint for building MEFMs for broad phenotype prediction, but specifically industrial metabolic engineering application. Results from MEFMs trained across domains of life will be used to refine strain designs, while the first designs from finetuning are being validated (vii).

6 Data sources and models ✗

Source code is located here: https://github.com/Mjvolk3/torchcell. The Neo4j Graph Database is served at UIUC's NCSA Radiant: https://torchcell-database.ncsa.illinois.edu/. Published open-source models will be made available at https://huggingface.co/. The team has access to computing resources at UIUC's NCSA and ANL's exascale Aurora supercomputer and other DOE-supported compute infrastructures. All AI models and experimental results will also be deposited into KBase/K-BERDL, providing a natural integration point with the Genesis Mission's AmSC/ModCon platforms.

7 Decision gate metrics ✗

At six months, we assess the quantitative metrics in Table 3 to gate technical readiness for months 6–9. Unmet criteria trigger the predefined mitigation strategy.

Metric 1 (Computational Scaling). Cross-organism transfer shows $\geq 5\times$ sample efficiency over from-scratch baselines (v) with transfer exponents > 0 and 95% CIs excluding zero demonstrating transfer (vi).

Mitigation. If unmet: Expand training data across phylogenetic gaps in TorchCell Database increasing overlap in multimodal inputs and phenotypes (i) and revise CGT architecture with a stricter approach in imposing physical constraints (iii).

Metric 2 (Experimental Scaling). RNA-barcoded YKOC data can be collected across ≥ 3 environmental conditions with perturbation-condition pairs for *I. orientalis* with data deposited in TorchCell Database.

Mitigation. If unmet: Reapply the published method with the multi-functional genome-wide CRISPR (MAGIC) system in *S. cerevisiae*. This mitigates the risk of applying the method to a new organism while providing an opportunity to develop an activation, interference, and deletion perturb-seq-style atlas for *S. cerevisiae* and to scale along a different axis – gene perturbation type.

8 Appendix 1: bibliography and references cited ✗

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